

2	(a) (i)	1. $pV = nRT$ $1.80 \times 10^{-3} \times 2.60 \times 10^5 = n \times 8.31 \times 297$ $n = 0.19 \text{ mol}$	C1 A1	[2]
		2. $\Delta q = mc\Delta T$ $95.0 = 0.190 \times 12.5 \times \Delta T$ $\Delta T = 40 \text{ K}$ <i>(allow 2 marks for correct answer with clear logic shown)</i>	B1 A1	[2]
	(ii)	$p/T = \text{constant}$ $(2.6 \times 10^5) / 297 = p / (297 + 40)$ $p = 2.95 \times 10^5 \text{ Pa}$	M1 A0	[1]
	(b)	change in internal energy is 120 J / 25 J internal energy decreases / ΔU is negative / kinetic energy of molecules decreases so temperature lower	B1 M1 A1	[3]

Page 3	Mark Scheme	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2013	9702	42